

Generalized Passive Face Liveness Detection Using Region-Specific Mixture-of-Experts

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Abstract—Face liveness detection is essential for ensuring authenticity in electronic Know Your Customer (eKYC) and face recognition systems, particularly to prevent unauthorized access via presentation attacks (PAs). Passive methods, which require no user interaction, offer user-friendly solutions but often struggle to generalize across diverse PA types and environmental conditions. To address this challenge, this paper proposes a theoretical modular passive liveness detection framework based on a Region-Specific Mixture-of-Experts (MoE) architecture, where each expert specializes in detecting specific liveness cues. The proposed framework is designed to include three experts, each focused on the eye region, skin texture, and environmental interaction. As a first step, this work presents the preliminary implementation and evaluation of the Skin Liveness Expert. The uniform LBP texture descriptors are extracted from skin regions and classified using various classifiers. On the CASIA-FASD dataset (in-domain), the best model Fully Connected Neural Network (FCNN) achieves an accuracy of 97.20%, in cross-dataset evaluation (CASIA → NUAA), it achieves an accuracy of 94.20%, indicating that region-specific texture-based cues transfer well. While the Eye Expert, Environmental Expert, and adaptive fusion mechanism remain under development, this paper establishes the foundation of a modular MoE framework with the potential to improve cross-dataset generalization and robustness against diverse attacks such as photos, replay videos, 3D masks, and deepfakes.

Keywords—Passive face liveness detection, Mixture-of-Experts, presentation attack detection, cross-dataset generalization, biometric security.

I. INTRODUCTION

The need for strong security frameworks has increased as biometric technologies, particularly facial recognition, are becoming integrated in sensitive areas like financial transactions, access control, and personal device security [1]. These systems are excellent at recognizing and authenticating people using their facial characteristics, but they are becoming more susceptible to spoofing or Presentation Attacks (PAs), in which attackers take advantage of easily accessible or fraudulent biometric data to obtain unauthorized access [2], [3]. These vulnerabilities pose significant risks, including financial fraud, identity theft, and breaches of privacy. Security measures such as Presentation Attack Detection (PAD) systems, which aim to identify and stop spoofing attacks, are vital in reducing these threats [4], [5]. Though face recognition systems are very good at identifying people based on facial features, they can't tell the difference between a real, live person (a genuine user) and a spoofing/presentation attacker (an imposter). PAD systems tackle this problem [6], [7]. Attackers frequently mimic authentic users by displaying a spoof face, a fake duplicate of a real user's face made using

printed pictures, replay videos, 2D, 3D masks, or cosmetically manipulated faces, or digitally manipulated images (e.g., deepfakes) [4]. The inability to differentiate between live and fake faces emphasizes the significance of face liveness detection as a critical security element in current face recognition systems, particularly those employed for access control and authentication [1].

A face liveness detection system is one of the PAD techniques, attempting to solve this issue by spotting PAs and guaranteeing that only live faces can access protected systems [1], [8], [9]. These algorithms can efficiently classify spoofing attempts when PAs are detected in conditions similar to those on which the system was trained [10]. However, such systems perform much worse in unconstrained conditions, new attack types, or cross-database settings, where environmental characteristics like illumination, camera quality, and human variety change from training circumstances [11], [12]. This restriction exposes a fundamental flaw in the reliability and scalability of current face liveness detection algorithms.

Existing face, or in general, liveness detection techniques are divided into two categories: active and passive [13]. Active methods need the user to do specified actions, such as blinking or nodding, to indicate their existence [14]. While successful, these solutions degrade the user experience and are not suitable for real-time or hands-free applications [13]. In contrast, passive methods may detect liveness from small facial and environmental cues alone, without requiring any interaction from the user [15]. While passive solutions are user-friendly and easy to implement, they often struggle in dynamic or unstructured environments [16]. They are especially vulnerable to sophisticated spoofing methods, such as AI-generated deepfakes and high-resolution 3D masks, which may provide very realistic and misleading biometric representations [17], [18]. As fraudsters use advanced technology, the fabricated identities they create become more realistic and complex, presenting substantial difficulties to the security and integrity of biometric systems. Traditional detection approaches, which were not intended to handle these developing threats, frequently fail, compromising the security and integrity of face recognition systems [8].

Despite significant advancements in passive face liveness detection technologies, current solutions face persistent challenges in balancing accuracy, real-time processing, and generalization, especially in the face of increasingly sophisticated PAs. These limitations have particular significance in uncontrolled environments, where factors such as lighting, camera quality, and subject diversity vary significantly from training conditions. Monolithic models often fail to generalize across such variability, leading to decreased robustness and reliability.

This research aims to address these challenges by proposing a theoretical and conceptual modular passive face liveness detection framework based on a region-specific Mixture-of-Experts (MoE) architecture, which specializes different facial regions and cues to particular experts to improve generalization. The primary aim of this research is to develop an optimised passive facial liveness detection system that effectively distinguishes live faces from spoofed ones, including complex attacks, like 3D masks and deepfakes, while maintaining computational efficiency.

The paper is structured into five sections. Section I introduces the research, detailing the context, aims, and objectives. Section II offers an examination of related research work in passive face liveness detection research. Section III describes the proposed approach, specifying the framework and design. Section IV presents the preliminary experimental results from the initial module implementation. Lastly, Section V concludes the paper with a summary of findings and future work.

II. RELATED WORK

This section discusses related research on passive facial liveness detection methods, emphasizing advancements from recent years. It examines studies that focus on PAD mechanisms to secure biometric systems against Facial PA, particularly through passive facial liveness detection techniques that do not require user interaction. The section summarizes research papers on passive facial liveness detection methods (non-AI-based and AI-based) and the dataset.

A. Passive Face Liveness Detection Techniques

Passive face liveness detection (PAD) aims to distinguish between live and spoofed faces without requiring user interaction. Traditional methods relied on handcrafted features like Local Binary Patterns (LBP) [10], [19], [20], motion cues [17], or frequency analysis and Fourier transforms [20], [21]. These techniques were often combined with threshold-based [10], [17], [20]–[23] or classical machine learning classifiers like SVMs [24]–[26], logistic regression [27], and other approaches [19].

With the rise of deep learning, CNN-based models became popular due to their ability to learn discriminative features directly from raw images [16], [18], [28]–[32]. Hybrid models combining CNNs and LSTMs [33], Hybrid models with other deep and machine learning models, ResNet, and SVM [34] or transfer learning [35], or lightweight architectures [36]–[38] such as LivenessNet [15], MobileNetV2 [39] have demonstrated improved accuracy.

However, these models often suffer from limited generalizability across datasets due to their monolithic design, which fails to adapt to varying spoofing conditions such as lighting, device quality, or attack type.

B. Mixture-of-Experts (MoE) Approaches

While most face liveness or PAs detection systems adopt monolithic models, a few studies have explored modular or expert-based strategies to improve generalization and robustness against diverse PAs conditions

One of the earliest works is by Kollreider et al. [40], who proposed a liveness verification system that integrates multiple handcrafted experts. Their system combines 3D motion cues and eye-blinking detection using optical flow and

velocity analysis, all without relying on specialized hardware. Though pioneering for its time, this approach is non-AI and relies on rigid rule-based thresholds, limiting adaptability to more sophisticated and unseen attacks.

In recent years, there has been a shift toward AI-based Mixture-of-Experts (MoE) architectures. Notably, Zhou et al. [41] introduced an Adaptive Mixture of Experts Learning (AMEL) framework, where each expert is trained on a specific dataset (e.g., CASIA, MSU) to capture domain-specific spoofing patterns. A Dynamic Expert Aggregation (DEA) module combines these expert outputs based on relevance to the test sample. Furthermore, a meta-learning strategy simulates unseen domains, enabling strong generalization across datasets. While AMEL achieves excellent performance (e.g., 11.31% HTER on unseen OULU), it assumes access to clearly separable domain-specific training data, and its relevance scoring may degrade if domain similarity is misestimated. Moreover, the study focuses only on physical PAs types, it doesn't consider digitally face-manipulated attack types.

Another study by Chen et al. [42] proposed MoAE-CR (Mixture-of-Attack-Experts with Class Regularization) for unified detection of both physical and digital face attacks. This model employs soft expert routing and class regularization modules to handle intra-class variance (e.g., various spoof types) and inter-class similarity (e.g., live vs. highly realistic attacks). Despite achieving state-of-the-art results on complex datasets like UniAttackData and JFSFDB, this architecture is computationally heavy and lacks analysis of real-time inference.

TABLE I. SUMMARY OF RELATED WORK

Ref.	Task	Expert Specialization	Fusion Method	Attack Types
Kollreider et al. [40]	Face Liveness Detection (PAD)	3D motion, eye-blink (rule-based)	Rule-based fusion	2D printed masks, replay attacks, photos
Zhou et al. [41]	Face Liveness Detection (PAD)	Domain-specific experts (per dataset)	Dynamic Expert Aggregation (DEA)	2D spoof attacks
Chen et al. [42]	Face Liveness Detection (PAD) + Forgery Detection (Facial manipulations)	Soft experts for each attack type	Soft routing + class regularization	Physical + Digital
ALFE [43]	Forgery Detection (Facial manipulations)	Forgery-specific (Deepfake, FaceSwap, etc.)	Adaptive knowledge distillation	Digital forgeries
Wang & Liu [44]	Forgery Detection (Deepfake)s	Capacity & scale on multi-scale face embeddings	Dynamic gating selects appropriate experts per input	Digital (Deepfake)
Essa [45]	Forgery Detection (Deepfake)	ViT encoders specialized in context (DaViT), frequency (iFormer), and resolution (GPViT)	MLP-Mixer feature fusion	DeepFake, Face2Face, FaceSwap, NeuralTextures, Celeb-DF
Our Work (Proposed)	Face Liveness Detection (PAD) + Forgery Detection (Facial manipulations)	Region-specific experts (eye, skin, background/environment)	Adaptive Fusion Module	2D prints and 3D masks, replay attacks, photos, deepfakes

A different perspective is offered by a study [43] on face forgery detection (digital face manipulations, not strictly liveness detection) that used Adaptive Learning From Experts (ALFE). Each expert is trained on a specific forgery type (e.g., Deepfakes, FaceSwap), and a student model learns to distill knowledge adaptively based on prediction confidence. Although effective for forgery classification, this work is task-specific and does not address live vs. spoof discrimination, which is central to PAD. Another study focused on digital PAs [44], the authors introduced the Deepfake Mixture of Experts (DFMoE) model, a Heterogeneous Mixture of Experts (HMoE) model. Their architecture employs a dynamic gating network to select and combine expert subnetworks of varying capacities and scales based on input characteristics. By coupling expert specialization with adaptive data augmentation, DFMoE enhances robustness against both known and unseen forgery types. Related direction utilizes feature fusion with specialized Vision Transformer (ViTs) encoders [45]. The authors employ three distinct ViTs, DaViT, iFormer, and GPViT, each focusing on capturing global context, frequency domain artifacts, and high-resolution details, respectively. These are merged via an MLP-Mixer fusion module, achieving state-of-the-art deepfake detection performance across FaceForensics++ and Celeb-DF benchmarks. While not using expert routing, this method demonstrates the power of modality-specialized feature processing in forgery detection.

Unlike previous works, as summarized in Table I, our proposed model introduces a conceptual modular framework, a region-specific MoE framework for passive face liveness detection. Instead of specializing experts by domain or attack type, which often limits flexibility when encountering new or mixed spoofing methods, we design experts that each focus on a specific visual cue, such as eye region, skin texture, or environmental consistency. This allows our model to capture diverse liveness indicators in parallel, enhancing generalization across unseen attacks, and reducing dependence on curated datasets.

III. METHODOLOGY AND SYSTEM DESIGN

This study proposes a novel conceptual modular, passive face liveness detection system using a region-specific MoE architecture. The architecture decomposes the detection task into three parallel expert modules, where the design of each expert module specializes in a specific visual or behavioral core module: *Eye Liveness expert*, *Skin Liveness expert*, and *Environmental expert*. This region-driven modularization approach was chosen to improve the system's ability to generalize across diverse face PA scenarios and environments. By isolating distinct tasks into specialized modules, the system can focus on specific abnormal features present in different spoofing methods, such as the lack of natural blinking in video attacks or unrealistic skin textures in mask-based attacks. This enables targeted detection strategies that increase the overall robustness of the system.

A. Architecture Overview

As illustrated in Figure 1, the system comprises three core components:

1. **Expert Modules:** Each expert is designed as a lightweight neural subnetwork that learns specialized features from its corresponding Region of Interest (ROI).

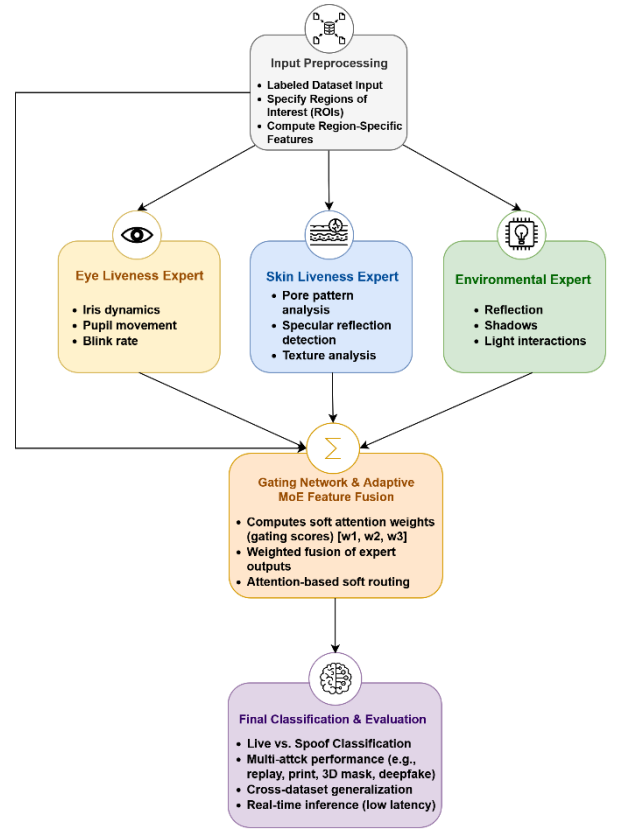


Figure 1. Overview of the proposed region-specific Mixture-of-Experts (MoE) framework for passive face liveness detection.

2. **Gating Network:** A high-level context-aware gating module would dynamically modulate the contribution weights of each expert's output based on input characteristics.
3. **Fusion & Classifier:** Expert outputs would be fused through a soft attention mechanism or gated summation, producing a final liveness score.

B. Expert Modules and Learning Objectives

Each expert learns to encode region-specific liveness cues by optimizing task-driven objectives over localized input features.

The *Eye Expert* learns temporal motion patterns using 3D convolutional blocks or stacked LSTMs/CNNs. Supervisory signals are driven by blink occurrence, gaze deviation, and inter-frame motion entropy. This module is particularly sensitive to replay attacks and static image spoofs.

The *Skin Expert* focuses on high-frequency texture descriptors (e.g., pores, specular reflections). It is trained using patch-level classification losses, texture regularization, and spatial smoothness constraints. This module focuses on detecting presentation artifacts in printed images and silicone masks.

The *Environmental Expert* analyzes global consistency between the facial foreground and the environmental background. Features include shadow coherence, background motion, and illumination dynamics. It uses regression and contrastive losses to model natural scene cues and detect compositional inconsistencies often present in digital or recombined inputs.

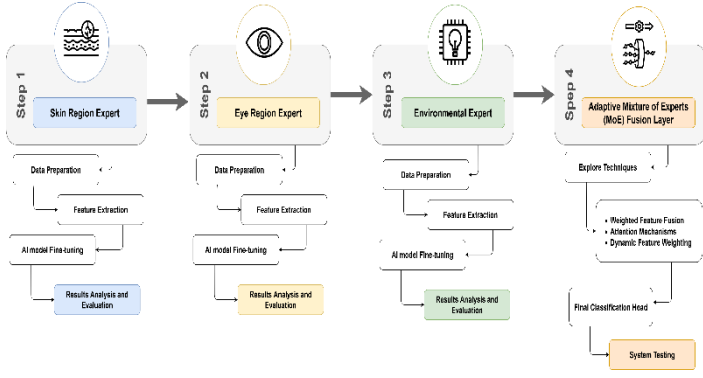


Figure 2 Stepwise development process for proposed region-specific Mixture-of-Experts (MoE) architecture for passive face liveness detection.

All experts operate independently and may share weights in initial convolutional layers to reduce model size.

C. Adaptive Fusion Mechanism

Unlike traditional MoE models that perform hard routing, we adopt a soft routing mechanism via a gating network. The gating module receives a global context vector (e.g., pooled features from the shared encoder or a separate attention summary) and outputs a soft weight vector $w=[w_1, w_2, w_3]$ via a softmax layer. These weights determine the contribution of each expert:

$$F_{fused} = \sum_{i=1}^3 w_i \cdot F_i \quad (1)$$

where F_i denotes the feature vector from the i^{th} expert, and w_i is its corresponding weight produced by the gating network. The gating module will be jointly trained with the experts using a combined loss function defined in Equation (2).

$$L_{total} = L_{cls} + \lambda \cdot L_{gate-reg} + \mu \cdot L_{expert-indep} \quad (2)$$

- L_{total} : binary cross-entropy loss for final liveness classification.
- $L_{gate-reg}$: optional entropy regularization to encourage confident gating decisions.
- $L_{expert-indep}$: optional penalty to encourage diversity in expert representations.

D. Input Assignment and Adaptivity

During training, each expert processes its designated region based on pre-defined ROIs. However, during inference, the gating module enables adaptive reliance on more confident experts. For instance, in low-resolution or occluded conditions, the gating module can learn to reduce the weight of the Eye Expert in favor of the Environmental module. This design will introduce a form of dynamic region prioritization, boosting robustness against partial occlusions, lighting changes, and attack diversity.

TABLE II. SUMMARY OF RESULTS

Classifier	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
SVM (Linear)	83.01	83.00	81.50	82.24
Random Forest	95.01	95.00	94.80	94.90
XGBoost	94.93	94.80	94.50	94.65
Logistic Regression	81.47	82.00	79.00	80.47
FCNN	97.20	97.10	97.00	97.05

IV. PRELIMINARY RESULTS AND DISCUSSION

A. Setup and Preliminary Results

As part of the ongoing development of the region-specific Mixture-of-Experts (MoE) framework, as illustrated in Figure 2, we have implemented and evaluated the Skin Liveness Expert module. This expert focuses on identifying presentation attacks by analyzing high-frequency skin texture descriptors and physiological inconsistencies. For this stage, we utilized the CASIA-FASD dataset for training, selecting relevant skin regions and applying preprocessing steps that included grayscale conversion and transformation into the YCbCr color space to enhance texture and chromatic features. Feature extraction was performed using Local Binary Patterns (LBP), and the extracted features were fed into various classifiers for evaluation, including Support Vector Machine (SVM), Random Forest, XGBoost, Logistic Regression, and a Fully Connected Neural Network (FCNN).

B. Discussion

From Table II, the FCNN model achieved the best performance, reaching an accuracy of 97.20%, precision of 97.10%, recall of 97.00%, and an F1-score of 97.05%. On the other hand, Random Forest and XGBoost are close to 95%, while SVM and Logistic Regression lag. This suggests that a small non-linear model works better with our LBP skin features than linear classifiers.

Moreover, the cross-dataset results of FCNN model (CASIA $\rightarrow$ NUAA) remains strong with accuracy 94.20%, Precision 94.35%, Recall 94.35%, and F1 94.14%, as illustrated in Table III. The findings illustrate the Skin Liveness Expert's ability in properly identifying fake skin textures in printed photos and video replay attacks. Furthermore, it demonstrates that concentrating on skin texture indicators retains valuable data despite changes in the

TABLE III. CROSS DATASET TESTING OF FNN MODEL

Train Test	→	Accuracy (%)	Precision (%)	Recall (%)	F1-Score (%)
CASIA $\rightarrow$ CASIA		97.20	97.10	97.00	97.05
CASIA $\rightarrow$ NUAA		94.20	94.35	94.35	94.14

dataset, so validating the effectiveness of region-specific analysis in passive liveness detection.

C. Limitations and Future Work

This is a preliminary, concept-oriented study. Only the Skin Expert has been evaluated, the Eye and Environmental experts, as well as the adaptive fusion process, have yet to be developed. The evaluation is limited to CASIA-FASD and NUAA, and we did not provide PAD-standard metrics (e.g., APCER, BPCER, ACER, EER) or runtime/latency. Therefore, our future work will extend the evaluation across multiple datasets (e.g., Replay-Attack, CelebA-Spoof, FaceForensics++), integrate the remaining experts with adaptive fusion, and analyze computational efficiency to validate the feasibility of real-world deployment.

V. CONCLUSION AND FUTURE WORK

This paper introduced a conceptual, modular passive face liveness detection framework based on a region-specific Mixture-of-Experts (MoE) architecture. By assigning distinct visual and behavioral cues to specialized expert modules, targeting the eye, skin, and environmental regions, the system aims to enhance generalization and robustness across diverse spoofing conditions. As of this work, only the *Skin Liveness Expert* has been implemented and evaluated. Preliminary results show effective detection of spoofed skin textures in print and replay attacks. The Eye and Environmental experts and the adaptive fusion layer remain under design. This proposal provides a solid foundation for a generalizable, modular system capable. Future work will focus on completing the full modular pipeline, benchmarking under cross-dataset and cross-attack settings and evaluating latency and model compression for real-time deployment.

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